Muhammad Hassaan Ibrahim

Working Student AI @ KLA

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Profile Overview

AI/ML engineer with **2+ years** of industry experience shipping LLM agents and production ML systems, currently building agentic AI for semiconductor manufacturing at KLA. M.Sc. at TU Dresden, graduating 11/2026.

Work Experience

KLA Corporation
Working Student – AI 08/2025 – Present | Dresden, Germany

Agentic AI – aiLITHO, a multi-agent LLM platform for chip manufacturing

- Gave semiconductor engineers **plain-language access to manufacturing data** by building tools and model integrations for aiLITHO, a production multi-agent LLM platform (LangGraph, LangChain).
- Made manuals, PDFs, slides, images and expert-interview recordings **searchable for the agents** with a multimodal ingestion pipeline, rebuilding all **66** knowledge modules with **100%** of search metadata preserved.
- Stopped agents guessing time ranges: a deterministic tool turns requests like “last 13 hours” into **exact query ranges**.
- Traced LLM evaluation cost (∼**662k input tokens**, about $3 per question) to oversized tool responses using MLflow and LLM-as-judge, and recommended bounded tool output.
- Integrated a self-hosted Gemma 4 mixture-of-experts LLM (131k context, vLLM) and contributed to model fine-tuning.
- Kept **478** unit tests passing with Docker and Azure DevOps CI.

Applied ML – wafer and microscope-image analytics

- **Automated CD-SEM wafer defect detection**, previously done by hand, by shipping a convolutional-autoencoder anomaly detector to production with Micron Technology (**15+ releases**).
- Gave engineers **a likely cause, not just a score**, with a module that groups flagged images by root cause.
- **Linked recurring wafer-map patterns to specific tools and process steps** with a root-cause analysis tool (PCA, autoencoders, SHAP, Plotly reports, 70 automated tests).
- Cut peak memory by ∼**42 GB** on a 4,632-wafer dataset by removing unneeded SHAP computations.
- **Caught inflated accuracy** from leakage between wafer lots with lot-held-out cross-validation, and reported results as descriptive rather than predictive.

NCT Dresden (National Center for Tumor Diseases)
Intern – Machine Learning 04/2025 – 03/2026 | Dresden, Germany

Vision-language pretraining for surgical video (12/2025 – 03/2026)

- Reproduced SurgLaVi’s narration-as-supervision pretraining from scratch, with no official code available: a pipeline over **94,094** clip–caption pairs from ∼1,100 surgical videos, and SurgCLIP, MedSigLIP and DINOv2 + LoRA encoders.
- Showed data quality beats volume: **64,962 filtered clips beat all 94,094** on zero-shot Cholec80 phase recognition (12.1% vs 8.6% F1) with 30% less data.
- Raised phase F1 from 12% to **44.4%** (∼4×) with an MS-TCN on the frozen features, showing they need temporal context.

Query-by-image video retrieval (04/2025 – 09/2025) · GitLab

- Finds the source video and timestamp of a single frame across 80 surgery videos at **98% Top-1 accuracy** with **sub-second** lookup (OpenCLIP, FAISS).
- Benchmarked general vs. surgical foundation-model encoders and three FAISS index types, stress-testing robustness on **8,000 queries** per perturbation.

NAECO Blue GmbH
Working Student – Data Scientist 06/2023 – 07/2024 | Bad Schwartau, Germany (Remote)

- Built **day-ahead energy-price forecasting** models for the German market, benchmarking statistical and deep learning approaches (**ARIMA**/**SARIMA**, **Prophet**, **XGBoost**, **N-BEATS**/**N-BEATSx**), and owned the full pipeline from **Selenium**/**BeautifulSoup** scraping through feature engineering to trained, evaluated models. *(details)*
- Delivered stakeholder-facing **Plotly Dash** and **Power BI** dashboards that made forecast output usable for non-technical decision-makers.
- Engaged directly with **customers** to gather requirements and feedback on forecast output, adjusting model behaviour and dashboard design to match how they actually used it.

Technical Skills

Languages	Python, C/C++, SQL, JavaScript, Bash
LLM & Agentic AI	LangGraph, LangChain, tool calling, multi-agent orchestration, MCP, RAG, BM25 and vector search (FAISS, pgvector), LLM evaluation (LLM-as-judge), vLLM, litellm, prompt engineering, Gemma, Qwen
ML / Deep Learning	PyTorch, TensorFlow/Keras, Hugging Face Transformers/Accelerate, scikit-learn, Optuna, SHAP, fine-tuning, LoRA, autoencoders, anomaly detection, diffusion LMs, vision-language models (CLIP, SigLIP)
Data & Documents	pandas, NumPy, SciPy, xarray, OpenCV, Pandoc, Docling, Plotly, Power BI, PostgreSQL
MLOps & DevOps	Docker / Docker Compose, Git, CI/CD (GitHub Actions, Azure DevOps), pytest, Ruff, uv, FastAPI, Pydantic, MLflow, Slurm, multi-GPU training (FSDP), CUDA, Linux / WSL2

Projects & Research

M.Sc. Thesis: Hybrid Autoregressive–Diffusion Architecture for Structured Code Generation 2026
TU Dresden (ScaDS.AI) · Supervisor: AFM Mohimenul Joaa · Referees: Prof. Dr. Michael Färber, Prof. Dr. Simon Razniewski

- Built a controlled comparison of **autoregressive** and **masked-diffusion** code generation: adapted **Qwen2.5-Coder-7B** twice on the same 7.4B-token corpus with the same training compute (4.5×10^{20} FLOPs), on **multi-node H100** clusters.
- Showed that diffusion's advantage depends on the information supplied rather than on the paradigm. Given edit locations and fix lengths, the diffusion model repaired **55.2%** of 310 real multi-location bugs (ConDefects) against 34.5% for AR infilling (**+20.6 points**), and the lead held against an infilling-trained baseline. Without given lengths, AR led by 11 points.
- Evaluated **six hybrid designs**, each against a matched AR control: compression of long inputs, candidate selection, judging, repair cascades, validators and shared-weight serving. Most gains disappear under fair controls; shared weights cut serving memory by **14%** while keeping AR outputs exact.
- Ran **pre-registered** studies with paired bootstrap intervals and exact tests on established benchmarks (HumanEval+, MBPP+, HumanEvalFix, CanItEdit, SAFIM, LiveCodeBench), reporting post hoc corrections and negative results transparently.
- Workshop paper from the results **in preparation**, targeting LLM4Code @ ICSE 2027.

TestForge: Agentic Test Generation for Python 2026
LangGraph, litellm, FastAPI, Docker, MLflow · github.com/hass44/testforge

- LLM agent that writes pytest suites for untested code that **kill 92.9% of mutants** across 16 real-world modules, via a self-correcting generate $\rightarrow$ repair $\rightarrow$ regenerate loop checked with AST analysis.
- Deployed as a FastAPI service with Docker-sandboxed test execution, provider-agnostic LLM access (Gemini, OpenAI, Hugging Face) and GitHub Actions CI/CD.

FirstSignal: Evidence-First AI Platform for Venture Diligence 2026
AI Hackathon, 3-person team · FastAPI, PostgreSQL/pgvector, LangGraph · live · GitHub

- Built and deployed the backend that **traces every claim to its source** and abstains when evidence is thin: multi-source ingestion (GitHub, web search, arXiv, PDF/DOCX) into vector search, with contradiction detection.
- Productionized with OpenAPI-generated clients, Alembic migrations, MLflow tracing (latency, tokens, cost) and a one-command Docker Compose stack.

MaxViT-Swin-UNet: Lymphocyte Segmentation 09/2023 – 07/2024
Pattern Recognition Lab, PIEAS · demo

- Designed a channel-boosted MaxViT-Swin-UNet that **beat UNet, Swin-UNet and MaxViT baselines** for state-of-the-art results on LYSTO, LYON'19 and Lizard, served to pathologists through a web app.

Education

TU Dresden	M.Sc. Computational Modelling and Simulation (Visual Computing) *Coursework:* Advanced ML, Large Language Models, ML for Robotics, Computer Vision, Data Engineering	10/2024 – exp. 11/2026
PIEAS, Islamabad	B.Sc. Computer Science	09/2020 – 2024

Additional

Leadership: Google Developer Student Clubs Campus Lead, PIEAS (08/2022 – 08/2023): led a 22-person core team and grew the chapter from **280 to 620** members through workshops, speaker sessions and partnerships with DataCamp and metaschool.

Languages: Urdu (native) English (fluent) German (A2)